

- 5(a) Magnetic flux density is defined to be the magnetic force acting per unit current and per unit length on a conducting wire [1]
placed at right angles to the direction of the magnetic field. [1]
- (b)(i) Direction of the magnetic flux density is *into* the plane of the page. [1]

(ii) Magnetic force on a charge particle, $F_B = Bqv \sin \theta$ [1]

$$\begin{aligned}
 &= (4.8 \times 10^{-3}) (1.6 \times 10^{-19}) (1.7 \times 10^7) (\sin 90^\circ) \\
 &= 1.3 \times 10^{-14} \text{ N}
 \end{aligned}
 \quad [1]$$

(iii) For circular motion, magnetic force provides for the circular motion,

$$\therefore F_B = \frac{mv^2}{r} \quad [1]$$

Therefore, the electron will move in a circular motion of radius,

$$\begin{aligned}
 r &= \frac{mv^2}{F_B} \\
 &= \frac{(9.11 \times 10^{-31}) (1.7 \times 10^7)^2}{1.3 \times 10^{-14}} = 0.020 \text{ m}
 \end{aligned}
 \quad [1]$$

Required distance, $d = 2r = 2(0.020) = 0.040 \text{ m}$ [1]

$$6(c)(i) \quad V_{\text{max}} = V_0 = 170 = 120 \text{ V} \quad [1]$$